ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

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Date: 5/1/2025
Compounds: Amoxicillin, Aspirin, Metformin, Omeprazole

New Compound Analysis

Name: Amospirocloxin
Formula: $C_{21}H_{30}N_4O_6S$
Weight: 462.56
Drug-likeness: 0.7

Activity:
Broad-spectrum antibacterial and anti-inflammatory

Target Analysis

Penicillin-binding proteins
Binding Affinity: -8.5
Inhibits bacterial growth

Cyclooxygenase (COX)
Binding Affinity: -9.2
Reduces inflammation and pain

H^+/K^+ ATPase
Binding Affinity: -7.5
Reduces gastric acid secretion

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Kidney, Gastrointestinal tract

Mitigation Suggestions:
- Co-administration with gastroprotective agents
- Regular liver function tests
- Hydration to support renal function

Pharmacokinetics

Absorption:
Moderate oral bioavailability

Distribution:
Wide tissue distribution

Metabolism:
Primarily hepatic with minor renal excretion

Excretion:
Renal and fecal

Half-life: 4-6 hours

Detailed Analysis

Amospirocloxin is a novel compound synthesized from the active constituents of Amoxicillin, Aspirin, Metformin, and Omeprazole. This broad-spectrum antibacterial and anti-inflammatory agent combines the cell wall synthesis inhibition from Amoxicillin, COX inhibition from Aspirin, and acid secretion reduction from Omeprazole. Although primarily metabolic processes for Metformin are not strongly integrated into the mechanism, its influence on blood glucose levels provides potential dual benefit when combined with these mechanisms. This multi-target approach increases its therapeutic index but presents moderate toxicity risks, particularly concerning the liver and kidneys. Considering the absorption, distribution, metabolism, and excretion profiles, as well as computational docking simulations, Amospirocloxin shows promising activity against infections with concurrent inflammatory and acidic conditions.
